RNA-seq analysis: bulk and single-cell

Part A: Drosophila KO vs WT (bulk) and PBMC clusters (scRNA-seq)

Part B: Wilk et al. 2020, single-cell atlas of severe COVID-19

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Part A: Analysis of RNA-seq data

Bulk RNA-seq (Rsubread + DESeq2) and single-cell RNA-seq (Seurat)

Bulk data: 7 fly samples aligned to chromosome X

5 KO · 2 WT

Drosophila melanogaster samples
(Sample1, 4–7 KO; 2–3 WT)

25,000

single-end 76 bp reads per sample
(Sample7: 22,500)

BDGP6.54

chrX FASTA + Ensembl GTF, as in the tutorial (DM_chrX.fa.gz, DM_chr.gtf.gz)

1. **`FASTQ`** 7 read files
2. **`buildindex()`** index chrX genome
3. **`subjunc()`** splice-aware alignment → BAM
4. **`featureCounts()`** reads per gene (GTF)
5. **`DESeq2`** normalise + test KO vs WT
6. **`IGV`** check reads on a DEG

Alignment: all reads mapped in all 7 samples

Code (Rsubread)

```
30 targets
31 dir.create("index")
32 dir.create("BAM")
33 buildindex(basename="index/chrX_melanogaster",reference="DM_chrX.fa.gz"
   ,gappedIndex = TRUE)
34 align(index="index/chrX_melanogaster",readfile1=targets$InputFile,input
   _format="FASTQ",output_format="BAM",output_file=targets$OutputFile,sort
   ReadsByCoordinates=TRUE)
35 propmapped(targets$OutputFile)
36 subjunc(index="index/chrX_melanogaster",readfile1=targets$InputFile,inp
   ut_format="FASTQ",output_format="BAM",output_file=targets$OutputFile,
37 sortReadsByCoordinates=TRUE)
```

subjunc is splice-aware: reads that span an intron are split across two exons. Each read is cut into 10 subreads; ≥ 3 must agree on the location (seed-and-vote).

Output: propmapped()

Sample	Total	Mapped	Unique	Multi
Sample1_KO	25,000	100%	24,959	41
Sample2_WT	25,000	100%	24,967	33
Sample3_WT	25,000	100%	24,957	43
Sample4_KO	25,000	100%	24,967	33
Sample5_KO	25,000	100%	24,961	39
Sample6_KO	25,000	100%	24,961	39
Sample7_KO	22,500	100%	22,462	38

Interpretation

100% mapped in every sample, almost all uniquely, 0 indels. QC note: Sample7's reads are the first 22,500 reads of Sample6, so it is a truncated copy, not an independent fly.

Counting: featureCounts assigns reads to genes

Code

```
fc <- featureCounts(files=targets$outputFile,annot.ext="DM_cnr.gtf.gz",
isGTFAnnotationFile=TRUE)
count2=fc$counts # Extract the counts
head(count2) # Print only the first 6 columns
anot=fc$annotation
head(anot)
count3=data.frame(anot$Length,count2)
colnames(count3)=c("width","Sample1","Sample2","Sample3","Sample4","Sam
ple5",
"Sample6", "Sample7")
head(count3, 7)
counts=count3[rowSums(count3[,-1])>=10,][,-1]
dim(counts)
```

- featureCounts reads the exon coordinates in the GTF and adds every read that overlaps an exon to that gene's gene_id
- Genes with < 10 reads in total are removed: they carry no information for testing

24 genes

kept after the ≥ 10-read filter (7 samples each)

~200 kb

region of chrX covered by the reads, so only a few dozen genes can be detected

Output = count matrix

Rows: FlyBase gene IDs (FBgn…). Columns: Sample1–7. Values: raw read counts.

Normalisation and DE method: DESeq2

Code

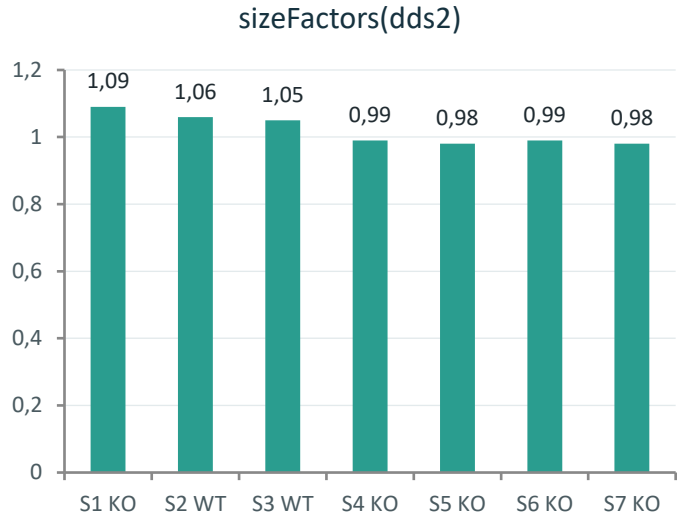
```
library(DESeq2)
treat=factor(c("KO","WT","WT","KO","KO","KO","KO"),levels=c("WT","KO"))
sampleinfo=data.frame(sampleName=colnames(counts),group=treat)
sampleinfo                          # check each sample has the right group
ddsMat <- DESeqDataSetFromMatrix(countData = counts,colData =
sampleinfo,design = ~ group)
dds2 <- DESeq(ddsMat)
sizeFactors(dds2)                   # normalization factors, for the slide
res <- results(dds2)
summary(res)
DEGs=data.frame(res[which(res$padj<0.05),])
print(DEGs,digits=2)
```

Normalisation

Median-of-ratios size factors correct for sequencing depth / library composition.

DE test

Negative binomial model per gene, Wald test KO vs WT, Benjamini–Hochberg padj.



All size factors are 0.98–1.09: libraries are very similar, so normalisation only makes small corrections (each count is divided by its sample's factor).

Result: 3 differentially expressed genes (padj < 0.05)

Code

```
res <- results(dds2)
summary(res)
```

```
DEGs=data.frame(res[which(res$padj<0.05),])
print(DEGs,digits=2)
```

Gene	baseMean	log2FC	Fold change	padj	In KO vs WT
Amun (FBgn0030328)	2722	+0.56	1.47×	0.024	▲ up
FBgn0030326	58	+0.61	1.53×	0.024	▲ up
FBgn0266421	440	−0.34	0.79×	0.024	▼ down

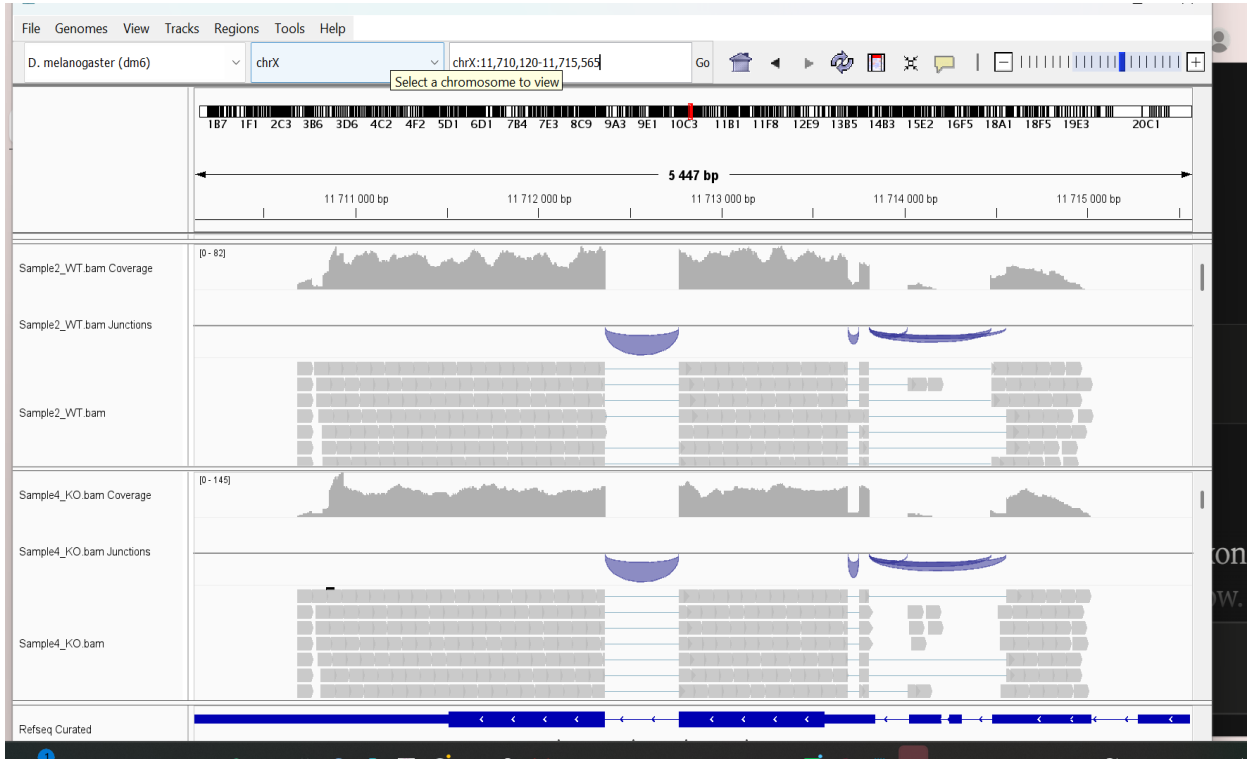
How to read it

log2FC 0.56 → 2^0.56 ≈ 1.47-fold. padj = p-value after correcting for 24 tests. 2 genes up, 1 down in KO (4 genes at padj < 0.1).

Caveat

Modest changes (< 2-fold), only 2 WT, and Sample7 is a copy of Sample6 (4 independent KO), so p-values may be optimistic.

IGV: reads on Amun are higher in KO than in WT



82

max coverage, WT (Sample2)

145

max coverage, KO (Sample4)

- Grey peaks = coverage on exons; gaps = introns
- Blue arcs = junction reads spliced across introns (subjunc)
- Same exon pattern in both, but higher KO coverage, consistent with DESeq2 (1.47×)

chrX:11,708,967–11,716,717 · dm6 · Amun is on the minus strand (arrows ←)

Single-cell RNA-seq: Seurat workflow

Data: sc_dataset1.rds, a Seurat object of blood cells (PBMCs), already log-normalised with the 2,000 most variable genes selected.

1 Scale

mean 0, variance 1 per gene

```
ScaleData(pbmc)
```

2 PCA

on the 2,000 variable genes

```
RunPCA(pbmc, features = VariableFeatures(pbmc))
```

3 SNN graph

shared nearest neighbours, 10 PCs

```
FindNeighbors(pbmc, dims = 1:10)
```

4 Clusters

Louvain; 0.3 → 4 clusters

```
FindClusters(pbmc, resolution = 0.3)
```

5 UMAP

2-D map of the same 10 PCs

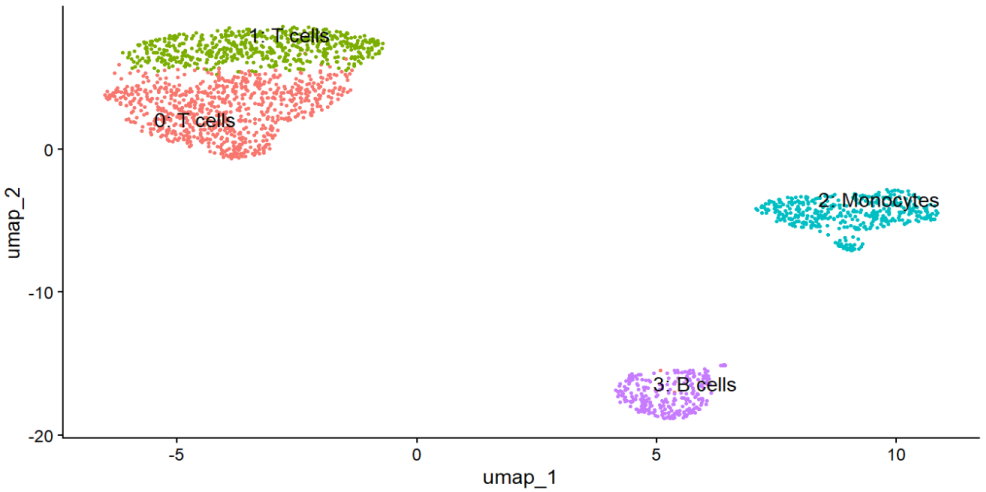
```
RunUMAP(pbmc, dims = 1:10)
```

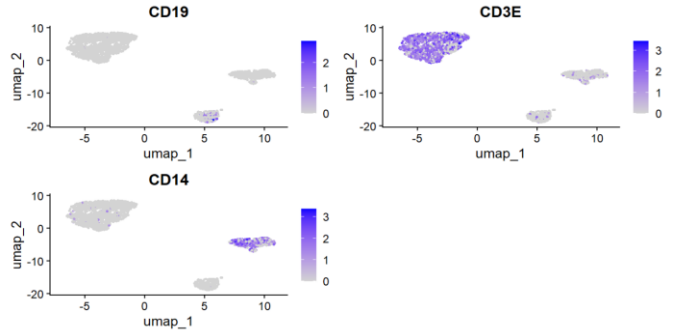
6 Annotate

marker genes → cell types

```
FeaturePlot(pbmc, c("CD19","CD3E","CD14"))
```

UMAP: 4 clusters = T cells, monocytes and B cells





Each dot is one cell; cells close together have similar expression. Clusters 0 and 1 touch → likely two T-cell subtypes. CD19 is patchy (low expression, dropout) but specific to cluster 3.

Cluster	Marker	Cell type
0, 1	CD3E	**T cells**
2	CD14	**Monocytes**
3	CD19	**B cells**

Part B: A single-cell atlas of severe COVID-19

Wilk AJ, Rustagi A, Zhao NQ, … Blish CA. Nature Medicine 26, 1070–1076 (2020)

Background and aim

1 · Biological background

Severe COVID-19 can cause acute respiratory distress syndrome (ARDS). A dysregulated immune response, often called a "cytokine storm", was suspected, but in early 2020 it was unclear which blood immune cells change and whether they produce the inflammatory cytokines.

2 · Aim / hypothesis

Build a single-cell atlas of the peripheral immune response in severe COVID-19, to find pathways that may cause immunopathology or give protective immunity, and test whether circulating immune cells drive the cytokine storm.

7 patients

hospitalised, 4 with ARDS on ventilators (8 samples)

6 controls

healthy donors

44,721

PBMCs sequenced (single cells)

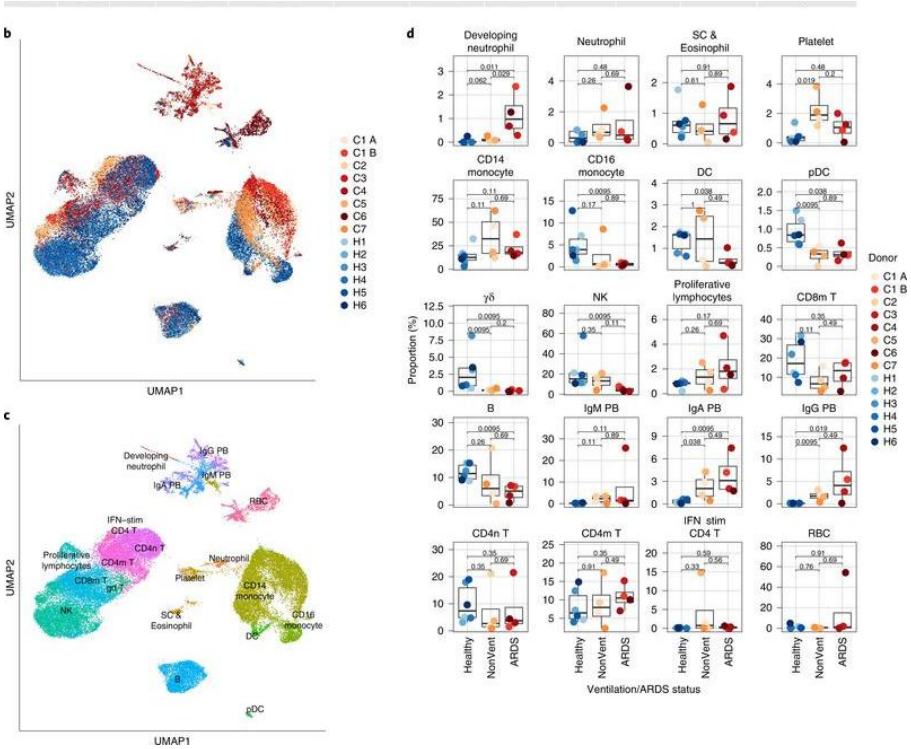
3 · Bioinformatic tools used in the paper

1	**Seq-Well + NovaSeq**	Portable, low-cost picowell scRNA-seq; Illumina sequencing (cell barcode + UMI)
2	**STAR + dropEst**	Align reads to GRCh37/hg19 (+ SARS-CoV-2); build the cell × gene UMI matrix
3	**QC filters**	Keep 1,000–15,000 UMIs, < 20% mito/rRNA; remove likely doublets; genes in ≥ 10 cells
4	**Seurat: SCTransform**	Normalisation with regression of quality covariates
5	**PCA → SNN → UMAP**	First 50 PCs; graph-based clustering → 30 clusters
6	**SingleR + DE genes**	Annotate clusters with a reference dataset; Wilcoxon tests; gene-module scores; pathway analysis

4 · Their pipeline vs. our course tutorial

Step	Course tutorial (Seurat PBMC)	Wilk et al. 2020
Raw data	10x data via Read10X()	Seq-Well; STAR + dropEst from FASTQ
QC	min.cells / min.features, % mito	UMI 1,000–15,000, < 20% mito/rRNA, doublet filter
Normalisation	NormalizeData (log-normalise)	SCTransform + covariate regression
PCs for graph/UMAP	10	50
Clusters	res. 0.3 → 4 clusters	30 clusters (many cell subtypes)
Annotation	By hand: CD3E, CD14, CD19	Top DE genes + SingleR reference
Comparison	One dataset, no groups	Patients vs controls: proportions, DE, module scores

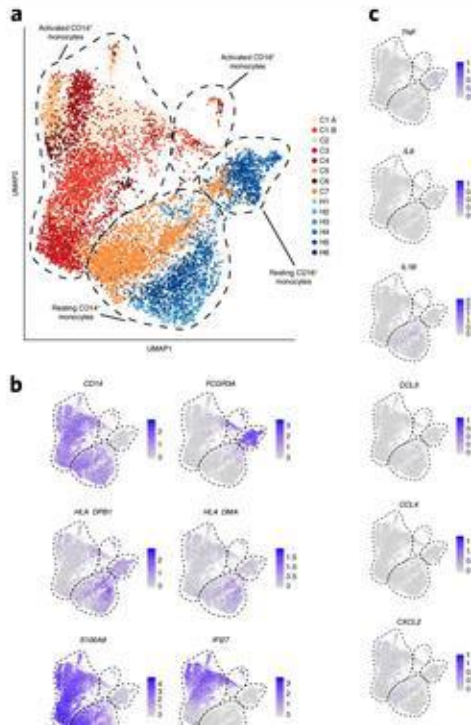
5 · Figure 1: UMAP of all 44,721 PBMCs



- 1a: patient table: 7 men (C1–C7), 4 ventilated with ARDS, 1 died
- 1b: UMAP coloured by donor: patient cells (red/orange) and healthy cells (blue) partly separate → disease changes cell states
- 1c: clusters annotated as cell types (T, NK, B, monocytes, DC, plasmablasts, developing neutrophils …)
- 1d: in ARDS patients, developing neutrophils and plasmablasts (IgA/IgG PB) increase, while CD16 monocytes, pDCs, NK and γδ T cells decrease

Source: Wilk et al. 2020, Nat Med, Fig. 1b–d

5 · Figure 2: monocytes lose HLA class II, not storm cytokines



Source: Wilk et al. 2020, Nat Med, Fig. 2a–c

- 2a: monocyte UMAP: patient cells (red/orange) sit in "activated" CD14 states, healthy cells (blue) in "resting" states
- 2b/2e/2f: patient CD14 monocytes have lower HLA class II (HLA-DPB1, HLA-DMA); module score healthy vs patients p = 0.0095 → weaker antigen presentation
- 2c: TNF, IL6, IL1B were hardly expressed by blood monocytes
- 2i–j: interferon-stimulated genes are up, but vary between patients
- **Conclusion: blood immune cells are reprogrammed, but are probably not the main source of the cytokine storm**

6 · Social impact and ethical considerations

Ethical use (benefits)

• Rapid, openly shared data (GEO, cellxgene) helped researchers worldwide during a pandemic
• Points to treatment targets (e.g. HLA class II loss, interferon response) rather than blanket anti-cytokine therapy
• Seq-Well is cheap and portable → studies possible in low-resource settings

Risks of unethical use

• Consent: critically ill, ventilated patients cannot consent themselves (proxy consent)
• Privacy: sequence data contain genetic variants → re-identification risk when data are open
• Speed vs rigour: preprint claims (plasmablast → neutrophil) were later disputed; over-hyped results can mislead treatment and media

My view: the benefits justified fast sharing, but open data need controlled access, and early findings must be clearly labelled as preliminary.

6 · Gender perspective: an all-male patient group

7 of 7

COVID-19 patients were men; sex of the controls is not reported

- Men had more severe COVID-19 in many cohorts; immunity differs by sex (X-linked immune genes such as TLR7, sex hormones)
- Findings such as HLA class II loss may therefore not apply equally to women: disease effect and sex effect cannot be separated
- Risk of a "male default": treatments developed from male data may work less well for women
- Gender (social) factors also matter: occupation, exposure risk and access to care differ between men and women

What should be done

Include and report both sexes (SAGER guidelines), analyse sex as a biological variable, and match controls by sex and age. This also applies to my own analyses: neither the fly samples nor the PBMC dataset report sex.

Summary

- Bulk: 100% mapped → 24 genes → DESeq2: 3 DEGs, e.g. Amun up 1.47× in KO, confirmed in IGV; QC: Sample7 is a copy of Sample6
- scRNA-seq: 4 clusters → T cells (CD3E), monocytes (CD14), B cells (CD19)
- Wilk 2020: blood immune cells reprogrammed in severe COVID-19; all-male cohort limits generalisation

References

- Wilk AJ, Rustagi A, Zhao NQ, et al. A single-cell atlas of the peripheral immune response in patients with severe COVID-19. Nat Med 26, 1070–1076 (2020). doi:10.1038/s41591-020-0944-y
- Alquicira-Hernandez J, Powell JE, Phan TG. No evidence that plasmablasts transdifferentiate into developing neutrophils in severe COVID-19 disease. Clinical & Translational Immunology (2021)
- Liao Y, Smyth GK, Shi W. The R package Rsubread is easier, faster, cheaper and better for alignment and quantification of RNA sequencing reads. Nucleic Acids Res 47, e47 (2019)
- Love MI, Huber W, Anders S. Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. Genome Biol 15, 550 (2014)
- Hao Y, et al. Integrated analysis of multimodal single-cell data (Seurat). Cell 184, 3573–3587 (2021)
- Heidari S, et al. Sex and Gender Equity in Research (SAGER guidelines). Res Integr Peer Rev 1, 2 (2016)